

1	Course Name:
	DATA MINING AND INFORMATION RETRIEVAL
2	Course Code:
	IQB10404
3	Year and Semester Offered:
	Year 1 Semester 2
4	Credit Hours Value:
	4 Credit Hours [4 SLT Credit]
5	Prerequisite(s), if applicable:
	NA
6	Course Synopsis:
	This course introduces principles techniques and practices of data mining and information retrieval. Students will explore methods to extract meaningful insights from large datasets and efficiently retrieve information from vast collection. Students will explore data preprocessing, classification, clustering, information retrieval (indexing and search engines) and text mining. The course emphasizes on practical applications, ethical considerations and development of critical thinking within the context of artificial intelligence (AI).
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Explain fundamental concepts and principles of data mining and information retrieval. 2. Construct data preprocessing, classification, and clustering techniques to analyze datasets for decision-making. 3. Implement indexing, text mining and pattern mining techniques for efficient information retrieval and text analysis.
8	Content Outline:
	Topic
	1. Introduction to Data Mining (DM) and Information Retrieval (IR)
	2. Data Preprocessing and Feature Selection
	3. Classification Techniques
	4. Clustering
	5. Information Retrieval Model and Web Search Engines (Indexing)
	6. Text Mining
	7. Pattern Mining and Dimensionality Reduction
9	References:
	Kotu, V., & Deshpande, B. (2014). Predictive analytics and data mining: concepts and practice with rapidminer. Morgan Kaufmann. (ISBN: 9780128014608)